

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s). * Rank 2 tensor (a).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ -M_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xy} = T_{pz}^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} M_{xxx} & 0 & 0 \\ -M_{xxx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -M_{xxx} & 0 \end{bmatrix}$$

$$M_{xxx} = M_{f2}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{yzx} & 0 & 0 \\ 0 & -M_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzx} = 3T_{du}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzx} & 0 & 0 \\ 0 & M_{yzx} & 0 \\ 0 & 0 & M_{xyz} \end{bmatrix}$$

$$M_{yzx} = -T_{du}^{(2)} + T_s^{(1)}$$

$$M_{xyz} = 2T_{du}^{(2)} + T_s^{(1)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxxx} & 0 \\ 0 & 0 & 0 & 0 & -M_{xxxx} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -M_{xxxx} \\ M_{xxxx} & -M_{xxxx} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -M_{xxxx} & 0 & 0 \end{bmatrix}$$

$$M_{xxxx} = M_{ga}^{(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxxx} & M_{xxxxy} \\ 0 & 0 & 0 & 0 & -M_{xxxx} & -M_{xxxxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{yzxz} & M_{xxxzx} \\ -M_{xxxzx} & M_{xxxzx} & 0 & -M_{yzxz} & 0 & 0 \\ -M_{xxxxy} & M_{xxxxy} & 0 & -M_{xxxzx} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{xxxx} &= -2T_{f1}^{(1)} \\
M_{xxxy} &= -2T_{faz}^{(1)} + 2T_{pz}^{(1)} \\
M_{yyzz} &= -4T_{faz}^{(1)} - T_{pz}^{(1)}
\end{aligned}$$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & M_{yyzz} & 0 \\ -M_{yyzz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yyzz} = -T_{pz}^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & M_{xxzx} & M_{xxxy} \\ 0 & -M_{xxzx} & M_{xxxy} \\ 0 & 0 & M_{zzxy} \\ 0 & M_{yyzx} & 0 \\ M_{yyzz} & 0 & 0 \\ M_{xxzx} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{xxxx} &= T_{f1}^{(2)} \\
M_{xxxy} &= -T_{faz}^{(2)} + T_{pz}^{(3)} \\
M_{zzxy} &= 2T_{faz}^{(2)} + T_{pz}^{(3)} + 2T_{pz}^{(4)} \\
M_{yyzz} &= -T_{faz}^{(2)} + T_{pz}^{(4)}
\end{aligned}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{yyzz} & M_{yyzy} \\ M_{yyzy} & -M_{yyzy} & 0 & M_{yyzz} & 0 & 0 \\ M_{xyyx} & M_{xyxx} & M_{xyzz} & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{yyzz} &= -T_{faz}^{(3)} + T_{pz}^{(6)} \\
M_{yyzy} &= T_{f1}^{(3)} \\
M_{xyxx} &= -T_{faz}^{(3)} + T_{pz}^{(5)} \\
M_{xyzz} &= 2T_{faz}^{(3)} + T_{pz}^{(5)} + 2T_{pz}^{(6)}
\end{aligned}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxzx} & M_{xxxy} & 0 & M_{xxxz} & M_{xxyx} \\ 0 & 0 & 0 & 0 & -M_{xxzx} & -M_{xxxy} & 0 & -M_{xxxz} & -M_{xxyx} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} & 0 & 0 & -M_{zzxy} \\ 0 & 0 & 0 & 0 & M_{yyzz} & M_{yyzy} & 0 & M_{yyzz} & M_{yyzy} \\ -M_{yyzy} & M_{yyzy} & 0 & -M_{yyzz} & 0 & 0 & -M_{yyzz} & 0 & 0 \\ -\frac{M_{xxxy}}{2} - \frac{M_{xxyx}}{2} & \frac{M_{xxxy}}{2} + \frac{M_{xxyx}}{2} & 0 & -M_{xxxz} & 0 & 0 & -M_{xxxz} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{xxxx} &= M_{ga}^{(1)} - 2T_{f1}^{(1)} + T_{f1}^{(2)} \\
M_{xxxy} &= -2T_{faz}^{(1)} - T_{faz}^{(2)} + 2T_{pz}^{(1)} + T_{pz}^{(3)} \\
M_{xxxz} &= M_{ga}^{(1)} - 2T_{f1}^{(1)} - T_{f1}^{(2)} \\
M_{xxyx} &= -2T_{faz}^{(1)} + T_{faz}^{(2)} + 2T_{pz}^{(1)} - T_{pz}^{(3)} \\
M_{zzxy} &= 2T_{faz}^{(2)} + T_{pz}^{(3)} + 2T_{pz}^{(4)} \\
M_{yyzz} &= -4T_{faz}^{(1)} - T_{faz}^{(2)} - T_{pz}^{(1)} + T_{pz}^{(4)} \\
M_{yyzy} &= -M_{ga}^{(1)} - 2T_{f1}^{(1)} \\
M_{yyxz} &= -4T_{faz}^{(1)} + T_{faz}^{(2)} - T_{pz}^{(1)} - T_{pz}^{(4)}
\end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{yzxz} & M_{yzxy} & 0 & M_{yzxz} & M_{yzxy} \\ M_{yzxy} & -M_{yzxy} & 0 & M_{yzxz} & 0 & 0 & M_{yzxz} & 0 & 0 \\ M_{xyxx} & M_{xyxx} & M_{xyzx} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzxz} = -T_{fz}^{(3)} - T_{pz}^{(2)} + T_{pz}^{(6)}$$

$$M_{yzxy} = T_{f1}^{(3)}$$

$$M_{yzxz} = -T_{fz}^{(3)} + T_{pz}^{(2)} + T_{pz}^{(6)}$$

$$M_{xyxx} = -T_{fz}^{(3)} + T_{pz}^{(5)}$$

$$M_{xyzx} = 2T_{fz}^{(3)} + T_{pz}^{(5)} + 2T_{pz}^{(6)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & 0 & M_{xxxx} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ -M_{xxxx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{xxxx} & 0 & 0 \\ 0 & 0 & -M_{xxxx} \\ 0 & -M_{xxxx} & 0 \end{bmatrix}$$

$$M_{xxxx} = M_{ga}^{(1)}$$